

Remove the Target. Preserve the Others.

Research focus: selective object removal when target and protected content interact.

The practical problem

Remove the specified instance and its effects.

Preserve nearby faces, text, structure and other instances.

A plausible image can still be an incorrect edit.

Our current stage

Editing infrastructure is implemented.

Strong-baseline validation is in progress.

A new method contribution remains to be demonstrated.

What We Have Built

Implemented pipeline; active denoising intervention remains a proposal.



Contract: target, operation and preservation requirements.

Controls: source-only masks and protected-region subtraction.

Final compositing preserves pixels outside the mask; it does not prove better generation.

Progress and Evidence

Verified September 16: one scene, five conditions, three seeds; not a broad benchmark.

Completed IEdit pilot

15/15 generations completed; 30 infrastructure tests passed.

Outside-mask raw L1: full 0.05176; masked 0.02246.

About 57% less pixel change, but signs still remain or regenerate.

Interpretation and status

Pixel preservation does not imply successful removal or preserved identity.

Native Fill removes rooftop signs but alters people.

ObjectClear imports successfully; GPU runs are still pending.

Where a Contribution Must Go Further

A contract, verifier or final composite alone is not the proposed contribution.

ObjectClear: target/effect localization and background fusion.

Consistency Critic: reference-guided consistency correction.

AdaEraser: adaptive attention suppression for removal.

Hypothesis: explicit instance ownership can reduce collateral damage when effects or boundaries overlap.

Sources: arxiv.org/abs/2505.22636v2 | arxiv.org/abs/2511.20614 | arxiv.org/abs/2605.15921

ObjectClear: What the Paper Does

Precise Object and Effect Removal with Adaptive Target-Aware Attention | CVPR 2026

Backbone and learning

SDXL-Inpainting with a CLIP visual object encoder.

Input: source image and target-object mask.

OBER supplies paired scenes and object-effect supervision.

Three mechanisms

ATA: learn where the target and its effects are.

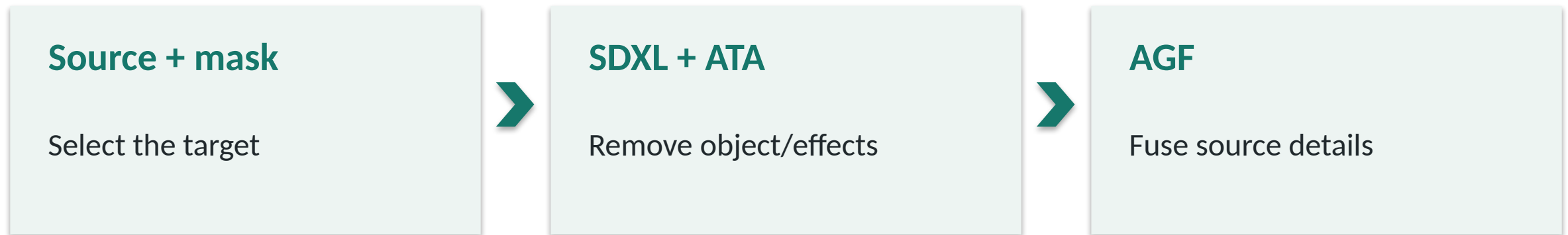
SVDS: vary denoising strength across regions.

AGF: use attention to fuse generated content with source details.

Zhao et al., arxiv.org/html/2505.22636v2 | Paper mechanisms, not our contributions

ObjectClear: Simplified Method Diagram

Conceptual redraw based on the paper; training supervision is distinct from inference input.



SVDS acts during denoising; AGF uses the predicted attention map.

OBER object-effect masks supervise training.

Source: arxiv.org/html/2505.22636v2, Sec. 3.2 | Simplified, not an exact computation graph

ObjectClear: Our Reproduction and Research Gap

Status checked September 16: submitted runs are pending; no reproduced quality claim yet.

Our reproduction work

Official code and FP16 weights prepared; pipeline import passes.

1044511: official sample, AGF on/off.

1044514: same ICDIT source/mask, 3 seeds x 2 modes; depends on smoke success.

Save attention and pre/post-fusion outputs.

What we still need to test

Does a strong remover already solve our failures?

Do simple protected masks suffice?

Candidate gap: protecting other instances when deletion regions interact.

An explicit protection controller remains a hypothesis.

Code caveat: AGF toggle changes early latent blending AND final fusion. Use within-run pre/post pairs to isolate final fusion.

Candidate Mechanism: Test Before Building

Proposed diagnostic, not a validated method or physical causal model.

1. Identify target, protected instances and uncertain regions.
2. At the same denoising state, vary target and protected reference evidence separately.
3. Test whether prediction responses distinguish target persistence from protected-content damage.
4. Build a regional controller only if this signal beats simple spatial heuristics.

Compare equal information and compute budgets. Report raw outputs before compositing.

Next Experiments and Decision Criteria

Use existing cases first; no new benchmark is required for this diagnostic phase.

Experimental sequence

Completed: ICDIT, 5 conditions x 3 seeds.

Next: validate ObjectClear and fusion effects.

Compare simple protection controls on held-out cases.

Then test the smallest intervention module.

Continue only if...

The failure persists on strong baselines.

Removal improves at comparable protection damage, or vice versa.

Gains survive raw-output and held-out evaluation.

Simple masks and equal-cost resampling cannot explain the gain.